

ARCHAEOLOGY



7. Industrial Revolution

1800–1900 AD

Let's travel back to the nineteenth century, when factories smoked and archaeologists uncovered lost worlds. As a boy, Heinrich Schliemann was enchanted by the stories of Homer, and with the Iliad in hand he found Troy on the hill of Hisarlik. In 1876, in Mycenae, he unearthed the gold "Mask of Agamemnon," though it is several centuries older than that king. The Rosetta Stone was found in 1799 by the army of Napoleon. It carried the same text in hieroglyphs, demotic script and Greek, so in 1822 Champollion was able to decipher the hieroglyphs. In the early nineteenth century Lord Elgin took sculptures from the Parthenon to London. In 1879 eight-year-old María noticed paintings of bison on the ceiling of the Altamira cave. In the 1840s Austen Henry Layard excavated the ruins of the Assyrian city of Nineveh. In 1856 the remains of a Neanderthal were found in the Neander Valley, and in 1863 in Pompeii Giuseppe Fiorelli poured plaster into hollows in the ash, recreating the shapes of Vesuvius's victims. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏴󠁧󠁢󠁥󠁮󠁧󠁿 As a boy, Heinrich Schliemann, the discoverer of Troy, was enchanted by the stories of...
 - a) Dante
 - b) Homer
 - c) Goethe
 - d) Shakespeare
2. 👁 In 1876 Schliemann discovered the gold funeral mask called the "Mask of Agamemnon" in...
 - a) Knossos
 - b) Troy
 - c) Athens
 - d) Mycenae
3. 📖 The Rosetta Stone allowed Champollion to decipher hieroglyphs (1822) because it also contained the same text in...
 - a) French
 - b) Arabic
 - c) Latin
 - d) Greek
4. 🇫🇷 The Rosetta Stone was found in 1799 by the army of...
 - a) Napoleon
 - b) Russia
 - c) Britain
 - d) Turkey
5. 🇬🇧 In the early 19th century Lord Elgin took to London sculptures from...
 - a) the Colosseum
 - b) the pyramids
 - c) the Parthenon
 - d) Stonehenge
6. 🇪🇸 In 1879 eight-year-old María, daughter of an amateur archaeologist, noticed in the Spanish cave of Altamira...
 - a) a golden treasure
 - b) a dinosaur skeleton
 - c) Roman coins
 - d) paintings of bison on the ceiling
7. ❤ In the 1840s Austen Henry Layard excavated the ruins of the Assyrian city of...
 - a) Mohenjo-daro
 - b) Nineveh
 - c) Carthage
 - d) Pompeii
8. 🌿 In 1856, in the Neander Valley in Germany, workers found the remains of...
 - a) a saber-toothed tiger
 - b) a dinosaur
 - c) a mammoth
 - d) a Neanderthal
9. ☆ In Pompeii (1863) Giuseppe Fiorelli poured plaster into hollows in the ash to recreate...
 - a) Roman coins
 - b) statues of gods
 - c) the street plans
 - d) the shapes of Vesuvius's victims

PHILOSOPHY

7. Industrial Revolution

1800–1900 AD



Let's move to the nineteenth century, when bold ideas were born alongside steam engines. The Dane Søren Kierkegaard wrote about anxiety, choice and individual responsibility, so he is considered a forerunner of existentialism. Arthur Schopenhauer, the philosopher of pessimism, held that life is above all suffering and desire, and that of all the arts music most deeply expresses the essence of the world. The bearded Karl Marx co-wrote "The Communist Manifesto" with Friedrich Engels in 1848. In 1882 Nietzsche proclaimed: "God is dead," speaking of the collapse of old values. The utilitarians Jeremy Bentham and John Stuart Mill believed that good is what brings the greatest happiness to the greatest number. In his essay "The Subjection of Women" from 1869, Mill called for equal rights for women and men. Henry David Thoreau lived alone for two years in a cabin by a pond called Walden. Hegel wrote that the owl of Minerva spreads its wings only at dusk, because wisdom comes after the events. Auguste Comte, founder of positivism, wanted knowledge to be based on facts and observation. Remember these ideas – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 The philosopher Arthur Schopenhauer believed that, of all the arts, the essence of the world is most deeply expressed by...
 - a) music
 - b) sculpture
 - c) architecture
 - d) poetry
2. 👁 The bearded philosopher who co-wrote "The Communist Manifesto" (1848) is...
 - a) Friedrich Nietzsche
 - b) John Stuart Mill
 - c) Immanuel Kant
 - d) Karl Marx
3. 🗨 In 1882 Nietzsche proclaimed the famous sentence: "God is..."
 - a) silent"
 - b) everywhere"
 - c) dead"
 - d) laughing"
4. ⚖ The utilitarians Jeremy Bentham and John Stuart Mill held that good is what brings...
 - a) obedience to the law
 - b) the greatest happiness to the greatest number
 - c) the greatest profit to the ruler
 - d) agreement with tradition
5. 🌲 For two years Henry David Thoreau lived alone in a cabin by a pond called...
 - a) Loch Ness
 - b) Walden
 - c) Baikal
 - d) Lake Geneva
6. 👤 In his essay "The Subjection of Women" (1869) John Stuart Mill called for...
 - a) equal rights for women and men
 - b) a return of the monarchy
 - c) the abolition of schools
 - d) a ban on newspapers
7. 💖 Søren Kierkegaard, who wrote about anxiety, choice and individual responsibility, is considered a forerunner of...
 - a) existentialism
 - b) positivism
 - c) Marxism
 - d) structuralism
8. 🌿 Auguste Comte, founder of positivism, believed knowledge should be based on...
 - a) revelation
 - b) dreams
 - c) facts and observation
 - d) tradition
9. ⭐ Schopenhauer, called the philosopher of pessimism, believed that life is above all...
 - a) an eternal game
 - b) suffering and desire
 - c) a struggle for fame
 - d) harmony and peace

HISTORY

7. Industrial Revolution

1800–1900 AD



Let's travel back to the stormy nineteenth century. In 1797 in Italy "Dąbrowski's Mazurka" was written, and it was sung by the Polish Legions fighting alongside Napoleon; it officially became Poland's anthem in 1927. In 1804 Napoleon Bonaparte crowned himself Emperor of the French in the cathedral of Notre-Dame in Paris. In 1812 his Grande Armée was defeated in Russia, where frost, hunger and disease killed most of the soldiers. The Congress of Vienna met in 1814–1815 to redraw the map of Europe after Napoleon. The emperor himself died in exile in 1821 on the island of Saint Helena. In 1810 Frédéric Chopin was born in the village of Żelazowa Wola. The November Uprising broke out in 1830 and the January Uprising in 1863 – 33 years apart. The year 1848 brought the Springtime of Nations, a wave of revolutions across Europe. From 1845 to 1849 Ireland suffered the Great Famine, caused by a disease of potatoes, the potato blight. In 1863 at Gettysburg Abraham Lincoln spoke of government of the people, by the people, for the people. Remember these events – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏰 In which village was Frédéric Chopin born?
 - a) Kraków
 - b) Duszniki-Zdrój
 - c) Żelazowa Wola
 - d) Poznań
2. 👁 In 1804 Napoleon Bonaparte crowned himself Emperor of the French in the cathedral of...
 - a) St. Peter's in Rome
 - b) Notre-Dame in Reims
 - c) Notre-Dame in Paris
 - d) St. Stephen's in Vienna
3. 🗨 The "Springtime of Nations" is the name for the wave of revolutions in Europe in...
 - a) 1848
 - b) 1863
 - c) 1815
 - d) 1871
4. 🇵🇱 The November Uprising broke out in 1830 and the January Uprising in 1863. How many years apart were they?

5. 🚩 In 1812 Napoleon's Grande Armée was defeated in...
 - a) Spain
 - b) Russia
 - c) England
 - d) Egypt
6. 🌐 The Congress of Vienna (1814–1815) met in order to...
 - a) divide up Africa
 - b) proclaim the end of slavery
 - c) found the United Nations
 - d) redraw the map of Europe after Napoleon
7. ❤ In the Gettysburg Address (1863) Abraham Lincoln spoke of government...
 - a) of the people, by the people, for the people
 - b) by the king and the nobility
 - c) with a strong hand
 - d) of the rich and the wise
8. 🍀 Ireland's Great Famine (1845–1849) was caused by a disease of...
 - a) potatoes (potato blight)
 - b) sea fish
 - c) sheep
 - d) wheat
9. ☆ Napoleon died in exile in 1821 on the island of...
 - a) Saint Helena
 - b) Malta
 - c) Corsica
 - d) Elba

1. c. 2. c. 3. a. 4. 33. 5. b. 6. d. 7. a. 8. a. 9. a

A N S W E R S



LINGUISTICS

7. Industrial Revolution

1800–1900 AD

Let's travel back to the nineteenth century, when languages and signs began to connect the world. L. L. Zamenhof, creator of Esperanto, published its first textbook in Warsaw in 1887 to help nations understand one another. Louis Braille developed an alphabet of raised dots read by touch; a full cell has 6 dots in two columns. In 1844 Samuel Morse sent the first telegraph message by tapping out dots and dashes. The Brothers Grimm, famous for fairy tales, were also linguists and began a great dictionary of the German language. Early in the century Samuel Linde published the first great "Dictionary of the Polish Language." After the January Uprising, schools in the Kingdom of Poland were forced to teach in Russian, and Poles answered with secret teaching. Scientists gave plants and animals two-part names, such as *Homo sapiens*, following the system of Linnaeus. And Oskar Kolberg spent his whole life collecting Polish folk songs and rituals. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎧 In 1887 L. L. Zamenhof, creator of Esperanto, published its first textbook in...
 - a) Vienna
 - b) Warsaw
 - c) London
 - d) Paris
2. 👁 The Braille alphabet, developed by Louis Braille, is made up of...
 - a) sounds
 - b) colored lines
 - c) hand gestures
 - d) raised dots
3. 🗨 The Brothers Grimm, known for their fairy tales, were also linguists and began writing a great...
 - a) atlas of the world
 - b) dictionary of the German language
 - c) Latin dictionary
 - d) Latin textbook
4. ⠠⠨⠠ How many dots does a full Braille cell have?

5. ⚡ In 1844 Samuel Morse sent the first telegraph message by tapping out dots and...
 - a) stars
 - b) circles
 - c) letters
 - d) dashes
6. 🇵🇱 After the January Uprising, schools in the Kingdom of Poland were forced to teach in...
 - a) Russian
 - b) Latin
 - c) French
 - d) German
7. ❤ In the early 19th century Samuel Linde published the first great...
 - a) atlas of Poland
 - b) collection of Polish proverbs
 - c) Latin grammar textbook
 - d) "Dictionary of the Polish Language"
8. 🌿 Many names of plants and animals have a two-part form, such as Homo sapiens, following the system of...
 - a) Darwin
 - b) Mendel
 - c) Pasteur
 - d) Linnaeus
9. ☆ Throughout his life Oskar Kolberg collected Polish...
 - a) postage stamps
 - b) folk songs and rituals
 - c) coins
 - d) butterflies

LITERARY STUDIES



7. Industrial Revolution

1800–1900 AD

Let's open the books of the nineteenth century, the age of great novels. In 1813, in "Pride and Prejudice," Jane Austen described how Elizabeth Bennet changes her mind about Mr. Darcy. In 1818 Mary Shelley created the scientist Frankenstein, who brought to life a creature made of body parts. In 1834 Adam Mickiewicz's "Pan Tadeusz" was published in Paris. In Dickens's "A Christmas Carol," the miser Scrooge is visited by three Christmas spirits. In Melville's "Moby-Dick," Captain Ahab hunts a white sperm whale that once bit off his leg. Longfellow's "The Song of Hiawatha" tells of a hero of the Native Americans. In Verne's novel, Captain Nemo traveled underwater in the submarine Nautilus. From 1887 on, detective Sherlock Holmes solved mysteries through deduction and observation. Bolesław Prus wrote "The Doll" about the merchant Stanisław Wokulski, and Henryk Sienkiewicz wrote the "Trilogy" to lift the hearts of Poles. Remember these characters – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏴 Longfellow's "The Song of Hiawatha" tells of a hero of...
 - a) the Vikings
 - b) the Native Americans
 - c) the Romans
 - d) the Chinese
2. 👁 In 1818 Mary Shelley wrote a novel about a scientist who brought to life a creature made of body parts. The scientist's name was...
 - a) Frankenstein
 - b) Nemo
 - c) Jekyll
 - d) Dracula
3. 🗨 In which year was Adam Mickiewicz's "Pan Tadeusz" published?
 - a) 1863
 - b) 1848
 - c) 1822
 - d) 1834
4. 🔍 Detective Sherlock Holmes solved mysteries through...
 - a) deduction and observation
 - b) chance
 - c) confessions
 - d) magic
5. 🚢 In Verne's novel, Captain Nemo traveled underwater in a submarine called the...
 - a) Bounty
 - b) Titanic
 - c) Nautilus
 - d) Victoria
6. 🧑 In Dickens's "A Christmas Carol," the miser Scrooge is visited by...
 - a) his dead parents
 - b) three Christmas spirits
 - c) the three kings
 - d) an angel and a devil
7. 💖 In "Pride and Prejudice" (1813) Jane Austen described how Elizabeth Bennet...
 - a) fights in a war
 - b) changes her mind about Mr. Darcy
 - c) becomes a queen
 - d) runs away to sea
8. 🍃 In Melville's "Moby-Dick," Captain Ahab hunts...
 - a) a shark
 - b) pirates
 - c) a white sperm whale
 - d) a giant squid
9. ☆ Bolesław Prus's novel "The Doll" tells the story of the merchant Stanisław...
 - a) Wokulski
 - b) Kmicic
 - c) Łęcki
 - d) Rzecki

CULTURE AND RELIGION

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, full of music, paintings and great exhibitions. The first World's Fair was held in London in 1851, in the Crystal Palace of glass and iron. The Eiffel Tower was built for the World's Fair in Paris in 1889. Stanisław Moniuszko, the father of Polish national opera, became most famous for his opera "Halka." In Vienna the waltz reigned thanks to Johann Strauss II, composer of "The Blue Danube." In 1874 Impressionism took its name from Monet's painting "Impression, Sunrise." In 1878 Jan Matejko painted "The Battle of Grunwald" to keep up Polish spirits under the partitions. Vincent van Gogh described his paintings in letters to his brother Theo, and he painted a series of sunflowers. And in 1858 St. Bernadette had apparitions of the Virgin Mary in Lourdes. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🇵🇱 **Stanisław Moniuszko is called the father of Polish national opera. His most famous opera is...**
 - a) "Carmen"
 - b) "Aida"
 - c) "Tannhäuser"
 - d) "Halka"
2. 🗼 **The Eiffel Tower was built for the World's Fair in Paris in...**
 - a) 1851
 - b) 1889
 - c) 1900
 - d) 1815
3. 🖼 **Impressionism took its name from Monet's painting "Impression,...**
 - a) Sunset"
 - b) Sunrise"
 - c) Water Lilies"
 - d) Cathedral"
4. 🌐 **The first World's Fair was held in London in 1851 in the...**
 - a) Buckingham Palace
 - b) Crystal Palace
 - c) Palace of Westminster
 - d) Tower of London
5. 🎵 **The Viennese waltz became popular in the 19th century thanks to the composer...**
 - a) Wagner
 - b) Johann Strauss II
 - c) Chopin
 - d) Beethoven
6. 🇵🇱 **Jan Matejko painted "The Battle of Grunwald" (1878) in order to...**
 - a) sell it to a museum in Berlin
 - b) keep up Polish spirits under the partitions
 - c) honor the reigning Russian tsar
 - d) advertise the city of Kraków
7. 💖 **In letters to his brother, Vincent van Gogh described his paintings. His brother's name was...**
 - a) Claude
 - b) Paul
 - c) Émile
 - d) Theo
8. 🌻 **Van Gogh painted a series of pictures of...**
 - a) roses
 - b) sunflowers
 - c) tulips
 - d) poppies in a vase
9. ⭐ **In 1858 St. Bernadette had apparitions of the Virgin Mary in...**
 - a) Lourdes
 - b) Fátima
 - c) Guadalupe
 - d) Częstochowa

ECONOMICS AND FINANCE

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when the modern market was being born. People still read Adam Smith, the economist who popularized the idea of the "invisible hand" of the market. Work in factories was timed by a factory whistle or siren, and workers often labored 12–14 hours a day. In 1840 the first postage stamp, the "Penny Black," appeared, showing Queen Victoria. In 1844 workers in Rochdale founded one of the first cooperatives. In 1848 the California Gold Rush began. Łódź grew into a great city thanks to its textile mills. Hipolit Cegielski from Poznań started out with a shop selling iron goods and tools, and later built a factory of farm machinery. Great operas and concerts were increasingly funded by middle-class ticket buyers. Alfred Nobel, the inventor of dynamite, left his fortune to fund prizes for outstanding achievements. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 In the 19th century great operas and concerts were increasingly funded by...
 - a) the composers themselves
 - b) the Church only
 - c) kings alone
 - d) middle-class ticket buyers
2. 👁 The first postage stamp, the "Penny Black" (1840), showed...
 - a) King George
 - b) the British lion
 - c) Napoleon
 - d) Queen Victoria
3. 🗨 The idea of the "invisible hand" of the market was popularized by the economist...
 - a) John Keynes
 - b) Karl Marx
 - c) Adam Smith
 - d) David Ricardo
4. 🏆 The California Gold Rush began in...
 - a) 1890
 - b) 1800
 - c) 1848
 - d) 1776
5. ⚙ In 19th-century factories, work was timed by...
 - a) a sundial
 - b) a bell on the town hall tower
 - c) the sunrise
 - d) a factory whistle or siren
6. 🧑 In Rochdale (1844) workers founded...
 - a) a central bank
 - b) the first stock exchange
 - c) an arms factory
 - d) one of the first cooperatives
7. ❤ Hipolit Cegielski, a businessman from Poznań, started out with a shop selling...
 - a) wine
 - b) fabrics
 - c) iron goods and tools
 - d) books
8. 🍃 In the 19th century Łódź grew into a large industrial city thanks to its...
 - a) textile mills
 - b) computer factories
 - c) car factories
 - d) aircraft plants
9. ⭐ Alfred Nobel, the inventor of dynamite, left his fortune to fund...
 - a) a church in Stockholm
 - b) the Swedish army
 - c) the expansion of his factory
 - d) prizes for outstanding achievements

PSYCHOLOGY

7. Industrial Revolution

1800–1900 AD



Let's look into the human mind in the nineteenth century. Ludwig van Beethoven composed masterpieces even though he was gradually losing his hearing; he wrote the Ninth Symphony almost completely deaf. In 1848 Phineas Gage survived an accident in which an iron rod flew through his head, and his personality changed greatly. In 1872 Charles Darwin studied how humans and animals express emotions. In 1879 in Leipzig Wilhelm Wundt founded the first laboratory of experimental psychology. Hermann Ebbinghaus studied memory by memorizing nonsense syllables. William James, the father of American psychology, was the brother of the writer Henry James. In a book from 1899 Sigmund Freud described the interpretation of dreams. Literature explored the soul too: Stevenson's Dr. Jekyll turned into the evil Mr. Hyde, and in "The Tell-Tale Heart" Edgar Allan Poe portrayed a tormenting sense of guilt. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 **Ludwig van Beethoven composed masterpieces even though he was gradually...**
 - a) unable to read
 - b) losing his hearing
 - c) losing his sight
 - d) forgetting how to read music
2. 👁️ **The first laboratory of experimental psychology was founded in Leipzig in 1879 by...**
 - a) Wilhelm Wundt
 - b) Ivan Pavlov
 - c) Carl Jung
 - d) Sigmund Freud
3. 📖 **In a book from 1899 (dated 1900) Sigmund Freud described the interpretation of...**
 - a) handwriting
 - b) faces
 - c) dreams
 - d) stars
4. 🧠 **Hermann Ebbinghaus studied memory by memorizing...**
 - a) poems
 - b) house numbers
 - c) faces
 - d) nonsense syllables
5. ⚡ **Phineas Gage survived an 1848 accident in which his head was pierced by...**
 - a) a cannonball
 - b) an iron rod
 - c) a stone from a sling
 - d) a lightning bolt
6. 👤 **William James, the father of American psychology, was the brother of the writer...**
 - a) Henry James
 - b) Jack London
 - c) Edgar Allan Poe
 - d) Mark Twain
7. 💖 **In Stevenson's novel (1886), Dr. Jekyll turns into...**
 - a) a werewolf
 - b) a vampire
 - c) a ghost
 - d) the evil Mr. Hyde
8. 🌿 **In a book from 1872 Charles Darwin studied how humans and animals...**
 - a) communicate in writing
 - b) count
 - c) sleep
 - d) express emotions
9. ⭐ **In the story "The Tell-Tale Heart," Edgar Allan Poe portrayed...**
 - a) the joy of wealth
 - b) a tormenting sense of guilt
 - c) a love of animals
 - d) a journey to the Moon

EDUCATION

7. Industrial Revolution

1800–1900 AD



Let's step into a nineteenth-century school. In 1837 Friedrich Fröbel founded the first kindergarten, where children played with blocks and sang. Pupils most often wrote on slates with a slate pencil, and the foundation of learning was the "three Rs": reading, writing and arithmetic. In the Russian partition, secret teaching was run by, among others, the Flying University. The "Sokół" Gymnastic Society, founded in 1867, combined physical exercise with patriotic education. For children, Maria Konopnicka wrote "Little Orphan Mary and the Gnomes." In 1860 Florence Nightingale opened in London the first secular school for nurses. In "Uncle Tom's Cabin" from 1852, Harriet Beecher Stowe described the fate of slaves in the USA. And Anne Sullivan, spelling letters with her finger on a palm, taught the deafblind Helen Keller to communicate. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏆 The first kindergarten was founded in 1837 by...
 - a) Friedrich Fröbel
 - b) John Amos Comenius
 - c) Maria Montessori
 - d) Janusz Korczak
2. 👁 In 19th-century schools, pupils most often wrote on...
 - a) tablets
 - b) slates with a slate pencil
 - c) wax tablets
 - d) papyrus
3. 🗨 In the Russian partition, secret teaching was run by, among others, the...
 - a) University of Vilnius
 - b) University of Warsaw
 - c) Flying University
 - d) Jagiellonian University
4. 📖 The foundation of 19th-century schooling was the "three Rs": reading, writing and...
 - a) religion
 - b) riding
 - c) arithmetic
 - d) running
5. 🦅 The "Sokół" (Falcon) Gymnastic Society (Lviv, 1867) combined physical exercise with...
 - a) trade
 - b) patriotic education
 - c) horse breeding
 - d) courtly dance lessons
6. 👧 For children, Maria Konopnicka wrote...
 - a) "The Doll"
 - b) "The Peasants"
 - c) "Pan Tadeusz"
 - d) "Little Orphan Mary and the Gnomes"
7. ❤ In 1860 Florence Nightingale founded in London the first secular school for...
 - a) village midwives
 - b) teachers
 - c) nurses
 - d) doctors
8. 🍃 In "Uncle Tom's Cabin" (1852) Harriet Beecher Stowe described...
 - a) life in China
 - b) the gardens of England
 - c) the fate of slaves in the USA
 - d) an expedition to the pole
9. ☆ Anne Sullivan taught the deafblind...
 - a) Emily Dickinson
 - b) Marie Curie
 - c) Jane Austen
 - d) Helen Keller

SOCIOLOGY

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when factories were changing whole societies. Industrial cities grew quickly because people moved from the countryside to the factories – we call this urbanization. The very word "sociology" was coined by Auguste Comte. From 1811 to 1816 the Luddites smashed weaving machines because they feared losing their jobs. Children worked in factories too, because they could get into narrow spaces in machines. In 1864 the tsar granted peasants ownership of their land in the Kingdom of Poland. In 1865, after the Civil War, slavery in the USA was finally abolished. After the fall of the Paris Commune in 1871, Eugène Pottier wrote the words of "The Internationale." In 1886 workers in Chicago fought for an 8-hour working day, which we remember on May 1. And in 1890 Jacob Riis used photographs to show the poverty of New York's slums. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 "The Internationale," the song of the workers' movement, was written after the fall of...
 - a) the Paris Commune (1871)
 - b) Napoleon (1815)
 - c) the Bastille (1789)
 - d) Rome (476)
2. 👁 In the 19th century industrial cities grew quickly because people...
 - a) moved from the countryside to the factories
 - b) looked for gold in the cities
 - c) fled from a war with Rome
 - d) had to live next to castles
3. 📖 The word "sociology" was coined by...
 - a) Auguste Comte
 - b) Émile Durkheim
 - c) Karl Marx
 - d) Max Weber
4. ⚙ The Luddites (1811–1816) smashed weaving machines because...
 - a) they wanted to steal parts
 - b) they could not operate them
 - c) they feared losing their jobs
 - d) the machines were ugly
5. 👦 Children also worked in 19th-century factories because...
 - a) they themselves wanted to
 - b) they could get into narrow spaces in machines
 - c) the law of the time required it
 - d) they were stronger than adults
6. 🇺🇸 Slavery in the USA was finally abolished in 1865 after...
 - a) the war with Mexico
 - b) the Civil War
 - c) the War of Independence
 - d) the Gold Rush
7. ❤ In 1886 workers in Chicago fought for...
 - a) a ban on machines
 - b) the abolition of taxes
 - c) an 8-hour working day
 - d) free housing
8. 🌿 The tsar granted peasants ownership of their land in the Kingdom of Poland in...
 - a) 1807
 - b) 1831
 - c) 1864
 - d) 1900
9. ☆ In "How the Other Half Lives" (1890) Jacob Riis used photographs to show...
 - a) the poverty of New York's slums
 - b) life at the queen's court
 - c) polar expeditions
 - d) Civil War battles

LAW

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when modern law was being born. In 1804 the Napoleonic Code appeared – a body of civil law. In 1807 the constitution of the Duchy of Warsaw was granted by Napoleon, abolishing the personal serfdom of peasants. In 1833 the British Empire banned slavery in its colonies. The Geneva Convention of 1864 protected wounded soldiers and medics. From 1871 Bismarck waged the Kulturkampf, the "culture struggle," against the Catholic Church. In 1872 the world's first national park, Yellowstone, was established in the USA. The Berne Convention of 1886 protects the rights of authors, such as composers, even abroad. In 1893 women first won the right to vote, in New Zealand. And in 1894 the French officer Alfred Dreyfus was wrongly convicted of treason. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏆 **The Berne Convention (1886) is an international agreement that protects...**
 - a) wild animals
 - b) workers' rights
 - c) national borders
 - d) the rights of authors, such as composers
2. 👁 **The Napoleonic Code of 1804 is a body of...**
 - a) maritime law
 - b) church law
 - c) civil law
 - d) military law
3. 🗨 **The Kulturkampf, German for "culture struggle," was Bismarck's campaign from 1871 against...**
 - a) the French army
 - b) the workers' movement
 - c) the Catholic Church
 - d) Austria
4. 🇵🇱 **The constitution of the Duchy of Warsaw was granted in 1807 by...**
 - a) Napoleon
 - b) the king of Prussia
 - c) Tsar Alexander I
 - d) Tadeusz Kościuszko
5. 🛡 **The Geneva Convention of 1864 protected...**
 - a) pets
 - b) freedom of trade
 - c) copyright
 - d) wounded soldiers and medics
6. 👩 **Women were the first in the world to win the right to vote in 1893 in...**
 - a) Poland
 - b) New Zealand
 - c) France
 - d) the USA
7. 💜 **In 1894 Alfred Dreyfus, a French officer, was...**
 - a) wrongly convicted of treason
 - b) promoted to general
 - c) elected president
 - d) declared a saint
8. 🌿 **The world's first national park, Yellowstone, was established in 1872 in...**
 - a) the USA
 - b) Poland
 - c) Switzerland
 - d) Canada
9. ⭐ **The British Empire banned slavery in its colonies in...**
 - a) 1776
 - b) 1901
 - c) 1815
 - d) 1833

OTHER (HUMANITIES)

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, full of novelties. Around 1826 Nicéphore Niépce made the first permanent photograph, "View from the Window at Le Gras." In 1828 the London Zoo was founded as a scientific center. Frédéric Chopin composed the "Revolutionary Étude" after hearing of the fall of the November Uprising. In "The Ugly Duckling," Hans Christian Andersen partly described his own childhood. In 1863 the Swiss Henri Dunant founded the Red Cross. The Eiffel Tower was completed in 1889, and 7 years later, in 1896, the first modern Olympic Games were held in Athens. Pierre de Coubertin believed that the most important thing in the Games is not winning but taking part and fair competition. In 1895 in Paris the Lumière brothers held the first public screening of films – the birth of cinema. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 Frédéric Chopin composed the "Revolutionary Étude" after hearing of the fall of...
 - a) Napoleon
 - b) the Constitution of 3 May
 - c) the November Uprising
 - d) the Springtime of Nations
2. 📷 The first permanent photograph ("View from the Window at Le Gras") was made around 1826 by...
 - a) Auguste Lumière
 - b) Louis Daguerre
 - c) George Eastman
 - d) Nicéphore Niépce
3. 🏅 The first modern Olympic Games were held in 1896 in...
 - a) London
 - b) Paris
 - c) Olympia
 - d) Athens
4. 🗼 The Eiffel Tower was completed in 1889. How many years later were the first modern Olympic Games held in Athens (1896)?

5. 🎬 In 1895 in Paris, the Lumière brothers held the first public screening of...
 - a) radio
 - b) films (cinema)
 - c) the phonograph
 - d) television
6. 🇨🇷 The Red Cross was founded in 1863 by the Swiss...
 - a) Alfred Nobel
 - b) Louis Pasteur
 - c) Henri Dunant
 - d) John Calvin
7. 💖 In the fairy tale "The Ugly Duckling," Hans Christian Andersen partly described...
 - a) the life of a king
 - b) his own childhood
 - c) a war with England
 - d) the history of Rome
8. 🐾 The first modern zoos opened in the 19th century. The London Zoo was founded as...
 - a) an animal market
 - b) a city slaughterhouse
 - c) a scientific center (1828)
 - d) a royal circus
9. ⭐ Pierre de Coubertin believed that the most important thing in the Games is not winning but...
 - a) the nation's glory
 - b) the medal and the prize
 - c) victory at any cost
 - d) taking part and fair competition

ARCHITECTURE AND URBAN PLANNING

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when cities grew both upward and outward. In 1851 the Crystal Palace rose in London, built mainly of cast iron and glass. Between 1853 and 1870 Baron Haussmann rebuilt Paris, laying out wide boulevards. In 1863 the world's first subway line opened in London. In 1875 the Palais Garnier opera house opened in Paris with its huge chandelier, and beneath the building lies an underground lake, a water tank. In 1882 construction began in Barcelona on the basilica of the Sagrada Família, where Antoni Gaudí worked for more than 40 years. In 1883 the Brooklyn Bridge connected Manhattan with Brooklyn. Tall buildings became practical thanks to the invention of the safety elevator, and in 1885 the first "skyscraper" was built in Chicago. In 1898 Ebenezer Howard proposed the "garden city," meaning towns surrounded by a green belt. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎪 **The Palais Garnier opera house in Paris (1875) has a huge chandelier, and beneath the building lies...**
 - a) a coal mine
 - b) an underground lake (a water tank)
 - c) a railway track
 - d) a stable for elephants
2. 👁 **London's Crystal Palace (1851) was built mainly of...**
 - a) marble
 - b) wood
 - c) cast iron and glass
 - d) brick and stone
3. 🗨 **The first "skyscraper" (the Home Insurance Building, 1885) was built in...**
 - a) Chicago
 - b) Warsaw
 - c) London
 - d) Paris
4. 🏗 **At the end of the 19th century tall buildings became practical thanks to the invention of...**
 - a) roof tiles
 - b) concrete
 - c) the safety elevator
 - d) brick
5. ⚡ **Between 1853 and 1870 Baron Haussmann rebuilt Paris by laying out...**
 - a) wide boulevards
 - b) Venetian-style canals
 - c) defensive walls
 - d) narrow lanes
6. 🌉 **The great suspension bridge opened in New York in 1883 connected Manhattan with...**
 - a) Brooklyn
 - b) Canada
 - c) Washington
 - d) Boston
7. ❤ **In 1882 construction began on the basilica of the Sagrada Família, where Antoni Gaudí worked, in...**
 - a) Rome
 - b) Madrid
 - c) Lisbon
 - d) Barcelona
8. 🌿 **In 1898 Ebenezer Howard proposed the "garden city," meaning...**
 - a) a castle with a park
 - b) a city under a glass dome
 - c) a garden on a factory roof
 - d) towns surrounded by a green belt
9. ⭐ **The world's first subway line opened in 1863 in...**
 - a) Berlin
 - b) New York
 - c) London
 - d) Paris



7. Industrial Revolution

1800–1900 AD

Let's travel back to the nineteenth century, when the world began to run on electricity. The unit of voltage, the volt, is named after Alessandro Volta, inventor of the battery. In 1827 Ohm's law was announced: current equals voltage divided by resistance. In 1831 Michael Faraday generated an electric current by moving a magnet in a coil of wire. In 1877 Thomas Edison recorded his voice on the phonograph, reciting "Mary Had a Little Lamb," and in 1879 he patented a practical light bulb with a carbon filament. Electric street lighting began replacing city lamps that burned gas. Nikola Tesla, who came from a Serbian family, together with George Westinghouse won the "war of the currents" against Edison by promoting alternating current. At the end of the century Tesla dreamed of transmitting electricity wirelessly around the whole world. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In 1877 Thomas Edison recorded his voice on the phonograph, reciting...
 - a) Shakespeare
 - b) the US national anthem
 - c) a letter to his wife
 - d) "Mary Had a Little Lamb"
2. 👁 The first practical light bulb with a carbon filament was patented in 1879 by...
 - a) Nikola Tesla
 - b) Alessandro Volta
 - c) Michael Faraday
 - d) Thomas Edison
3. 🔊 The unit of voltage, the volt, is named after the inventor of the battery...
 - a) Alessandro Volta
 - b) Georg Ohm
 - c) André-Marie Ampère
 - d) James Watt
4. ⚡ Ohm's law (1827) says that current equals voltage divided by...
 - a) mass
 - b) power
 - c) time
 - d) resistance
5. ⚡ In 1831 Michael Faraday generated an electric current by moving...
 - a) a cat across amber
 - b) a magnet in a coil of wire
 - c) a ball across a table
 - d) an oar in water
6. 🧑 Nikola Tesla and George Westinghouse won the "war of the currents" against Edison by promoting...
 - a) static electricity
 - b) direct current
 - c) alternating current
 - d) battery current
7. ❤ Nikola Tesla, creator of the AC motor, came from a family that was...
 - a) English
 - b) American
 - c) German
 - d) Serbian
8. 🌿 The first electric street lighting began replacing city lamps that burned...
 - a) gas
 - b) beeswax candles
 - c) olive oil
 - d) kerosene on carts
9. ☆ At the end of the 19th century Nikola Tesla dreamed of transmitting electricity...
 - a) by carrier pigeon
 - b) in barrels on carts
 - c) only through water pipes
 - d) wirelessly around the whole world

COMPUTER SCIENCE AND TELECOMMUNICATIONS

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when modern communication and calculating machines were born. In 1804 the Jacquard loom was controlled by punched cards. Around 1843 Ada Lovelace wrote the first algorithm for the Analytical Engine of Charles Babbage, which was never fully built because there was not enough money or precision. In Morse code, the SOS signal is three dots, three dashes, three dots. In 1858 the first transatlantic telegraph cable connected Ireland with Newfoundland. In 1876 Alexander Graham Bell patented the telephone. In 1887 Heinrich Hertz detected the radio waves that Maxwell had predicted. For the 1890 US census, Herman Hollerith built a tabulating machine that used punched cards. And at about age 21 Guglielmo Marconi began sending signals without wires, by radio. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In 1876 Alexander Graham Bell patented a device that transmitted the voice over a distance: the...
 - a) gramophone
 - b) telephone
 - c) radio
 - d) telegraph
2. 👁 The Jacquard loom (1804) was controlled by...
 - a) sounds
 - b) punched cards
 - c) pedals
 - d) a keyboard
3. 📄 Around 1843 Ada Lovelace wrote the first algorithm for the Analytical Engine of...
 - a) Thomas Edison
 - b) Blaise Pascal
 - c) Charles Babbage
 - d) Alan Turing
4. ⚡ In Morse code, the SOS signal is...
 - a) three dashes
 - b) three dots, three dashes, three dots
 - c) a dot and a dash
 - d) six dots
5. 🗂 The machine that tabulated the 1890 US census on punched cards was built by...
 - a) Charles Babbage
 - b) Samuel Morse
 - c) Herman Hollerith
 - d) Henry Ford
6. 🌐 The first transatlantic telegraph cable (1858) connected...
 - a) England with India
 - b) France with Brazil
 - c) Ireland with Newfoundland
 - d) Spain with Cuba
7. 💖 At about age 21 Guglielmo Marconi began sending signals...
 - a) by pigeons
 - b) through underground cables
 - c) without wires (by radio)
 - d) with lamps
8. 🌿 Radio waves, whose existence Maxwell had predicted, were detected experimentally in 1887 by...
 - a) Isaac Newton
 - b) Charles Darwin
 - c) Louis Pasteur
 - d) Heinrich Hertz
9. ☆ Babbage's Analytical Engine was never fully built because...
 - a) there was not enough money or precision
 - b) it burned in a factory fire
 - c) the Church banned it
 - d) no one knew how to power it

1. b 2. b 3. c 4. b 5. c 6. c 7. c 8. d 9. a

A N S W E R S

BIOMEDICAL ENGINEERING

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when doctors got new tools. In 1816 René Laennec invented the stethoscope; at first it was a rolled-up tube of paper. The name comes from the Greek words "stethos" and "skochein," meaning chest and to look. The first hearing aids were trumpets held up to the ear. In 1846 the first general anesthesia with ether during a public operation was demonstrated in Boston. In 1851 Hermann von Helmholtz invented the ophthalmoscope, which lets doctors look at the inside of the eye. In 1853 Alexander Wood popularized the syringe with a fine needle for hypodermic injections. In 1867 Thomas Allbutt built a short clinical thermometer that measured fever in 5 minutes. In 1895 Wilhelm Röntgen took the first X-ray image, showing his wife's hand with her wedding ring. And in 1896 Scipione Riva-Rocci developed a blood pressure device with an arm cuff. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **René Laennec invented the stethoscope in 1816; at first it was a rolled-up tube of...**
 - a) a leaf
 - b) a leather strap
 - c) metal
 - d) paper
2. 👁 **In 1895 Wilhelm Röntgen took the first X-ray image, which showed...**
 - a) a horse's leg
 - b) a cat's skull
 - c) his own heart
 - d) his wife's hand with her wedding ring
3. 🗨 **"Stethoscope" comes from the Greek words "stethos" and "skopein," meaning...**
 - a) heart and to hear
 - b) chest and to look
 - c) body and to measure
 - d) voice and to write
4. 🌡 **The short clinical thermometer that measured fever in 5 minutes was built in 1867 by...**
 - a) Galileo
 - b) Isaac Newton
 - c) Anders Celsius
 - d) Thomas Allbutt
5. 🩺 **The blood pressure device with an arm cuff was developed in 1896 by...**
 - a) Louis Pasteur
 - b) Joseph Lister
 - c) Scipione Riva-Rocci
 - d) Robert Koch
6. 👤 **The syringe with a fine needle for hypodermic injections was popularized in 1853 by...**
 - a) Florence Nightingale
 - b) Ignaz Semmelweis
 - c) Edward Jenner
 - d) Alexander Wood
7. 💖 **The first hearing aids of the 19th century were...**
 - a) magnets worn around the neck
 - b) implants
 - c) microchips
 - d) trumpets held up to the ear
8. 🍃 **The ophthalmoscope, invented by Hermann von Helmholtz (1851), lets doctors look at...**
 - a) the teeth
 - b) the inside of the eye
 - c) the inside of the ear
 - d) the heart
9. ⭐ **The first general anesthesia with ether during a public operation (1846) was demonstrated in...**
 - a) Paris
 - b) Kraków
 - c) Boston
 - d) Vienna

CHEMICAL ENGINEERING



7. Industrial Revolution

1800–1900 AD

Let's travel back to the nineteenth century, when chemistry entered the factories. In 1823 Charles Macintosh produced a waterproof fabric using rubber. In 1828 Friedrich Wöhler astonished chemists by producing urea, a compound from living organisms, in the laboratory. In 1839 Charles Goodyear hardened rubber with sulfur – this process is called vulcanization. In 1853 Ignacy Łukasiewicz obtained kerosene through the distillation of crude oil. In 1856 18-year-old William Perkin discovered by accident the purple mauveine, the first synthetic dye. In 1861 the Solvay process made it possible to mass-produce soda ash for making glass and soap. In 1867 Alfred Nobel patented dynamite. In 1869 Dmitri Mendeleev arranged the elements in a table according to atomic mass. And in 1899 the company Bayer put aspirin on the market. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏆 The first synthetic dye, the purple mauveine, was discovered by accident in 1856 by...
 - a) 18-year-old William Perkin
 - b) Louis Pasteur
 - c) Alfred Nobel
 - d) Marie Curie
2. 👁 In 1867 Alfred Nobel patented...
 - a) dynamite
 - b) gunpowder
 - c) nylon
 - d) gasoline
3. 🗨 In 1839 Charles Goodyear hardened rubber with sulfur. This process is called...
 - a) refining
 - b) distillation
 - c) pasteurization
 - d) vulcanization
4. ⚖ In 1869 Dmitri Mendeleev arranged the elements in a table according to...
 - a) price
 - b) date of discovery
 - c) atomic mass
 - d) color
5. ⚡ In 1853 Ignacy Łukasiewicz obtained kerosene by carrying out...
 - a) distillation of crude oil
 - b) melting of wax
 - c) grinding of coal
 - d) fermentation of grapes
6. 🧑 In the Solvay process (1861) made it possible to mass-produce soda ash needed to make...
 - a) gold
 - b) glass and soap
 - c) bread
 - d) cigarettes
7. ❤ In 1828 Friedrich Wöhler astonished chemists by producing in the laboratory...
 - a) water from fire
 - b) urea, a compound from living organisms
 - c) gold from lead
 - d) diamond from coal
8. 🌿 In 1823 Charles Macintosh produced a waterproof fabric using...
 - a) rubber
 - b) beeswax
 - c) tar
 - d) whale fat
9. ☆ Aspirin (acetylsalicylic acid) was put on the market in 1899 by the company...
 - a) Nokia
 - b) Bayer
 - c) Nestlé
 - d) Ford

CIVIL ENGINEERING AND TRANSPORT

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when travel picked up speed. In 1819 the first steamship crossed the Atlantic, though it covered most of the distance under sail. In 1829 George Stephenson's locomotive, the Rocket, won the contest for the Liverpool–Manchester line. In 1842 the first railway line on Polish lands opened, running from Wrocław to Oława. In 1869 the Suez Canal connected the Mediterranean Sea with the Red Sea. On the roads rode the penny-farthing, or high-wheeler, with pedals on a huge front wheel. In 1886 Karl Benz patented a car with an internal combustion engine, and in 1888 Bertha Benz drove it about 100 km to show the world that the car worked. In 1889 the Eiffel Tower rose, and it remained the tallest structure in the world until 1930. The Trans-Siberian Railway was built from 1891 to 1916 – 25 years. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **George Stephenson's first locomotive for the Liverpool–Manchester line (1829) was called...**
 - a) the Lightning
 - b) the Arrow
 - c) the Eagle
 - d) the Rocket
2. 🗺️ **The Suez Canal (1869) connected the Mediterranean Sea with...**
 - a) the Baltic Sea
 - b) the Black Sea
 - c) the Atlantic
 - d) the Red Sea
3. 🗨️ **The first railway line on Polish lands (1842) ran from Wrocław to...**
 - a) Gdańsk
 - b) Oława
 - c) Kraków
 - d) Warsaw
4. 🚊 **Construction of the Trans-Siberian Railway began in 1891, and the main route was completed in 1916. How many years did construction take?**

5. 🚲 **The bicycle with pedals on a huge front wheel was called...**
 - a) a railway handcar
 - b) a kick scooter
 - c) a penny-farthing (high-wheeler)
 - d) Drais's running machine
6. 🚢 **In 1886 Karl Benz patented...**
 - a) a steamship
 - b) a car with an internal combustion engine
 - c) a horse-drawn tram
 - d) an airship
7. 💖 **In 1888 Bertha Benz drove a car about 100 km in order to...**
 - a) escape from the police
 - b) deliver a letter
 - c) show the world that the car worked
 - d) win a race
8. 🍃 **The first steamship crossed the Atlantic in 1819, but it covered most of the distance...**
 - a) underwater
 - b) under sail
 - c) under oars
 - d) under tow
9. ⭐ **The Eiffel Tower was the tallest structure in the world until...**
 - a) 1970
 - b) 1890
 - c) 1930
 - d) 1900

MATERIALS ENGINEERING

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, a century of new materials. In 1824 Joseph Aspdin patented a cement named after the limestone of the Isle of Portland. In 1856 Henry Bessemer invented a method for cheaply producing steel. Reinforced concrete appeared: concrete strengthened with steel bars. Cast-iron frames began to be built into pianos because they withstand greater string tension. Aluminum was so expensive that Napoleon III had cutlery made of it for his guests of honor. Celluloid, one of the first plastics, replaced ivory in billiard balls. From the mid-century paper was made from wood instead of rags, so books and newspapers became cheap. At first the filament of Edison's light bulbs was made from carbonized bamboo. And the Eiffel Tower was built of wrought, or puddled, iron. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the 19th century cast-iron frames began to be built into pianos because...
 - a) they withstand greater string tension
 - b) they are lighter than wood
 - c) they muffle the sound
 - d) they shine beautifully
2. 👁 In 1856 Henry Bessemer invented a method for cheaply producing...
 - a) rubber
 - b) paper
 - c) steel
 - d) glass
3. 📄 The cement patented in 1824 by Joseph Aspdin was named after the limestone of the Isle of...
 - a) Jersey
 - b) Wight
 - c) Portland
 - d) Man
4. 🏗 Reinforced concrete is concrete strengthened with...
 - a) steel bars
 - b) glass
 - c) wood
 - d) straw
5. 🔄 Celluloid, one of the first plastics, replaced ivory in...
 - a) teeth
 - b) billiard balls
 - c) uniform buttons
 - d) cutlery
6. 🧑 In the 19th century aluminum was so expensive that Napoleon III...
 - a) banned its mining
 - b) ordered coins minted from it
 - c) had an imperial crown made of it
 - d) had cutlery made of it for his guests of honor
7. 💖 Paper made from wood instead of rags, used from the mid-19th century, meant that...
 - a) paper became stronger
 - b) Europe's forests disappeared
 - c) people went back to parchment
 - d) books and newspapers became cheap
8. 🍃 At first, the filament of Edison's light bulbs was made from carbonized...
 - a) gold
 - b) bamboo
 - c) wool
 - d) glass
9. ⭐ The Eiffel Tower was built of...
 - a) aluminum
 - b) wrought (puddled) iron
 - c) stainless steel
 - d) concrete

1 a 2 c 3 c 4 a 5 b 6 d 7 d 8 b 9 b

ANSWERS

MECHANICAL ENGINEERING

7. Industrial Revolution

1800–1900 AD



Let's travel back to the age of steam and machines. James Watt improved the steam engine by adding a separate steam condenser, and he compared his engines with the work of horses – hence "horsepower," a unit of power. The spinning machine and the power loom transformed the textile industry. In 1804 Richard Trevithick built the first steam locomotive running on rails. In 1851 Isaac Singer's sewing machine was powered by a foot treadle. On American farms windmills were used mainly for pumping water. Music boxes and barrel organs played tunes thanks to a pinned cylinder or punched holes. Between 1891 and 1896 Otto Lilienthal made over 2,000 flights in a glider. And in 1893 Rudolf Diesel built an engine in which the fuel is ignited by hot compressed air. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Music boxes and barrel organs played tunes thanks to...**
 - a) people hidden inside
 - b) a pinned cylinder or punched holes
 - c) electric current
 - d) water
2. 👁 **James Watt improved the steam engine by adding...**
 - a) a sail
 - b) a water wheel
 - c) an electric motor
 - d) a separate steam condenser
3. 🗨 **The unit "hp," which James Watt devised to compare engines with horses, measures...**
 - a) mass
 - b) length
 - c) speed
 - d) power
4. ⚙ **In 1893 Rudolf Diesel built an engine in which the fuel is ignited by...**
 - a) a match
 - b) a spark plug
 - c) hot compressed air
 - d) sunlight
5. 🧵 **Isaac Singer's sewing machine (1851) was powered by...**
 - a) steam
 - b) electricity
 - c) wind
 - d) a foot treadle
6. 🧶 **The spinning machine and the power loom transformed the...**
 - a) food industry
 - b) textile industry
 - c) computer industry
 - d) aviation industry
7. ❤ **In 1804 Richard Trevithick built the first...**
 - a) rocket
 - b) elevator
 - c) steam locomotive running on rails
 - d) typewriter
8. 🌿 **Windmills on 19th-century American farms were used mainly for...**
 - a) heating
 - b) pumping water
 - c) powering cities
 - d) grinding coffee
9. ⭐ **Between 1891 and 1896 Otto Lilienthal, an aviation pioneer, made over 2,000 flights in a...**
 - a) powered airplane
 - b) glider
 - c) rocket
 - d) balloon

7. Industrial Revolution

1800–1900 AD



Let's go down into the mines of the nineteenth century. Coal formed millions of years ago from the remains of plants. Miners took canaries underground because the birds warned of poisonous gas, and Davy's lamp of 1815 protected against explosions of methane. Miners have their own bands and celebrate St. Barbara's Day, Barbórka, honoring their patron. In 1854 Łukasiewicz's oil mine, the first in the world, was established in Bóbrka. That same year John Snow stopped a cholera epidemic in London by ordering the handle of a contaminated pump removed. The Great Stink of 1858 pushed London's authorities to build a sewer system. In 1882 the world's first hydroelectric power plant started in the USA, in Appleton. And smog and factory smoke in England changed the color of moths to dark. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In many countries miners have their own bands and celebrate...
 - a) St. Andrew's Eve
 - b) Maritime Day
 - c) St. Barbara's Day (Barbórka)
 - d) Midsummer Eve
2. 👁 Miners took canaries into the mines because the birds...
 - a) warned of poisonous gas
 - b) sang to raise spirits
 - c) caught insects
 - d) found coal
3. 🗨 Davy's miner's lamp (1815) protected against explosions of...
 - a) sulfur
 - b) water
 - c) coal
 - d) methane
4. 🚧 London's Great Stink (1858) pushed the authorities to build...
 - a) bridges
 - b) a sewer system
 - c) factories
 - d) a new palace
5. ⚡ Łukasiewicz's oil mine (1854), the first in the world, was established in...
 - a) Texas
 - b) Łódź
 - c) Bóbrka
 - d) Baku
6. 🧑 In 1854 John Snow stopped a cholera epidemic in London by ordering...
 - a) the handle of a contaminated pump removed
 - b) a ban on selling fish
 - c) all the city's theaters closed
 - d) infected houses burned down
7. ❤ The world's first hydroelectric power plant was started in 1882 in...
 - a) Poland
 - b) China
 - c) Egypt
 - d) the USA (Appleton)
8. 🌿 Coal formed millions of years ago from...
 - a) sand
 - b) the remains of plants
 - c) dinosaur bones
 - d) lava
9. ☆ Smog and factory smoke in 19th-century England changed the color of moths to...
 - a) dark
 - b) transparent
 - c) white
 - d) red

1. c 2. a 3. d 4. b 5. c 6. a 7. d 8. b 9. a

A N S W E R S

ASTRONOMY



7. Industrial Revolution

1800–1900 AD

Let's look at the sky of the nineteenth century. Back in 1781 Uranus was discovered by William Herschel, who was a musician by profession and built his own telescopes. On January 1, 1801, Giuseppe Piazzi discovered Ceres, the first asteroid. Joseph von Fraunhofer discovered dark lines in the spectrum of the Sun. In 1833 the Leonid meteor shower was so intense that many thought the end of the world had come. In 1838 Friedrich Bessel was the first to measure the distance to a star. In 1840 John William Draper took the first photograph of the Moon. In 1846 Neptune was discovered thanks to mathematical calculations. In 1847 Maria Mitchell became the first American woman to discover a comet. And in 1877 Giovanni Schiaparelli saw "canali" on Mars, wrongly understood as canals built by Martians. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The planet Uranus was discovered in 1781 by William Herschel, who by profession was a...
 - a) musician
 - b) doctor
 - c) soldier
 - d) baker
2. 👁 Neptune was discovered in 1846 thanks to...
 - a) chance
 - b) a space telescope
 - c) a legend
 - d) mathematical calculations
3. 📖 The first asteroid (today a dwarf planet), Ceres, was discovered on January 1, 1801, by...
 - a) Galileo
 - b) Giuseppe Piazzi
 - c) Copernicus
 - d) Kepler
4. ⚖ In 1838 Friedrich Bessel was the first to measure...
 - a) the mass of the Moon
 - b) the temperature of the Sun
 - c) the age of the Earth
 - d) the distance to a star
5. 🌌 In 1877 Giovanni Schiaparelli saw "canali" on Mars, which was wrongly understood to mean...
 - a) seas
 - b) forests
 - c) rivers of lava
 - d) canals built by Martians
6. 👩 In 1847 Maria Mitchell became the first American woman to discover...
 - a) a comet
 - b) a planet
 - c) a black hole
 - d) a galaxy
7. 💖 The first photograph of the Moon (a daguerreotype) was taken in 1840 by...
 - a) John William Draper
 - b) Copernicus
 - c) Newton
 - d) Galileo
8. 🌿 In the spectrum of the Sun, Joseph von Fraunhofer discovered...
 - a) two suns
 - b) red dots
 - c) dark lines
 - d) a green ray
9. 🌟 The Leonid meteor shower of 1833 was so intense that many people thought...
 - a) it was fireworks
 - b) the end of the world had come
 - c) the sun had exploded
 - d) the stars had gone out

MATHEMATICS



7. Industrial Revolution

1800–1900 AD

Let's travel back to the nineteenth century, a golden age of mathematics. The story goes that young Gauss was supposed to quickly add the numbers from 1 to 100, and he answered 5050. In 1822 Joseph Fourier showed that complex waves, such as sounds, can be broken down into simple sine waves. Évariste Galois wrote down his most important discoveries on the night before the duel in which he was killed. In 1854 George Boole created an algebra in which variables take only two values: true or false. In 1858 the Möbius strip was described – a strip of paper joined with a single twist, which has only one side. As a statistician, Florence Nightingale devised a chart called the "rose," a polar area diagram. In 1880 the Venn diagram represented sets as overlapping circles. In 1889 Sofia Kovalevskaya became one of the first women in the world to be a professor of mathematics. And Georg Cantor showed that there are different kinds of infinity. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Joseph Fourier (1822) showed that complex waves, such as sounds, can be broken down into...**
 - a) prime numbers
 - b) nothing but squares
 - c) colors
 - d) simple sine waves
2. 👁 **A Venn diagram (1880) represents sets as...**
 - a) pyramids
 - b) lines
 - c) overlapping circles
 - d) columns of numbers
3. 🗨 **George Boole (1854) created an algebra in which variables take only two values:**
 - a) even or odd
 - b) big or small
 - c) true or false
 - d) plus or minus
4. ➦ **Young Gauss was supposed to quickly add up the numbers from 1 to 100. What answer did he give?**

5. 🌀 **A Möbius strip (1858) is a strip of paper joined with a single twist, which has...**
 - a) only one side
 - b) the shape of a square
 - c) four sides
 - d) three edges
6. 👩 **In 1889 Sofia Kovalevskaya became one of the first women in the world to be a...**
 - a) president
 - b) general
 - c) queen
 - d) professor of mathematics
7. 💖 **Évariste Galois wrote down his most important discoveries on the night before...**
 - a) a voyage to America
 - b) the duel in which he was killed
 - c) an exam
 - d) his wedding
8. 🍃 **As a statistician, Florence Nightingale devised a chart called...**
 - a) the network chart
 - b) the heat map
 - c) the "rose" (a polar area diagram)
 - d) the 3D bar chart
9. ☆ **At the end of the 19th century Georg Cantor showed that there are...**
 - a) negative numbers
 - b) different kinds of infinity
 - c) square numbers
 - d) fractions

OTHER (TECHNOLOGY)



7. Industrial Revolution

1800–1900 AD

Let's travel back to the nineteenth century, full of inventions for everyday life. Early in the century Napoleon offered a prize for a way to preserve food, and Nicolas Appert won it by preserving food in airtight jars. Canned food was also invented, so that food would not spoil on expeditions. Louis Braille lost his sight as a child after an accident in his father's workshop. When the first railways appeared, some doctors feared that fast travel would harm passengers' health. In 1849 Walter Hunt supposedly invented the safety pin in three hours to pay off a debt. In 1873 Levi Strauss and Jacob Davis patented riveted jeans. From 1874 the company Remington sold a typewriter with the QWERTY layout. In 1887 Emile Berliner invented the gramophone, which played sound from a flat disc. And in 1888 George Eastman launched the Kodak camera with the slogan: "You press the button, we do the rest." Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In 1887 Emile Berliner invented the gramophone, which played sound from...
 - a) a tape
 - b) a flat disc
 - c) a crystal
 - d) a wax cylinder
2. 👁 In 1888 George Eastman launched the Kodak camera with the slogan: "You press the button,..."
 - a) you are the artist"
 - b) you get the picture"
 - c) we do the rest"
 - d) the world smiles"
3. 🗨 The first practical typewriter with the QWERTY layout was sold from 1874 by the company...
 - a) Ford
 - b) Remington
 - c) Kodak
 - d) Singer
4. 🍲 Canned food was invented at the beginning of the 19th century so that...
 - a) city water would be cleaner
 - b) animals had fodder for winter
 - c) cooking dinner would be easier
 - d) food would not spoil on expeditions
5. ⚡ The safety pin (1849) was supposedly invented in three hours, by twisting wire to pay off a debt, by...
 - a) Karl Benz
 - b) Thomas Edison
 - c) Levi Strauss
 - d) Walter Hunt
6. 🧑 In 1873 Levi Strauss and Jacob Davis patented...
 - a) the umbrella
 - b) sneakers
 - c) the cowboy hat
 - d) riveted jeans
7. ❤ Louis Braille lost his sight as a child after an accident in...
 - a) his father's workshop
 - b) the forest
 - c) school
 - d) a mine
8. 🌿 Nicolas Appert won Napoleon's prize by preserving food in...
 - a) brine
 - b) snow
 - c) honey
 - d) airtight jars
9. ☆ When the first railways appeared, some doctors feared that fast travel would...
 - a) anger the ancient gods
 - b) harm passengers' health
 - c) stop the passage of time
 - d) dry up rivers and lakes

BIOLOGY

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when biology was uncovering the secrets of life. In 1838–1839 Schleiden and Schwann announced the cell theory: organisms are made up of cells. From 1831 to 1836 Charles Darwin sailed around the world aboard the ship "Beagle." On the Galápagos Islands he noticed that the finches differed mainly in the shape of their beaks. For about 20 years he delayed publishing his theory because he feared how society would react, until Alfred Russel Wallace, studying animals in the islands of the Malay Archipelago, independently reached similar conclusions. In 1859 Darwin published "On the Origin of Species." The theory caused great disputes because it held that humans and apes share common ancestors. In 1861 the feathered first bird Archaeopteryx was found in Germany. Gregor Mendel studied the laws of heredity by crossing peas. And Darwin even tested whether earthworms respond to sounds by playing to them on the piano and the bassoon. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 Charles Darwin tested whether earthworms respond to sounds by playing to them on...
 - a) the violin and the drum
 - b) the trumpet
 - c) the guitar
 - d) the piano and the bassoon
2. 👁 Darwin noticed that the finches of the Galápagos Islands differed mainly in the shape of their...
 - a) feathers
 - b) eyes
 - c) claws
 - d) beaks
3. 🚢 Darwin sailed around the world aboard the ship...
 - a) "Beagle"
 - b) "Endeavour"
 - c) "Victoria"
 - d) "Titanic"
4. ⚖ Gregor Mendel studied the laws of heredity by crossing...
 - a) mice
 - b) peas
 - c) roses
 - d) fruit flies
5. ⚡ Independently of Darwin, Alfred Russel Wallace arrived at the theory of evolution by studying animals in...
 - a) the Alps
 - b) the North Pole
 - c) the islands of the Malay Archipelago
 - d) the Sahara
6. 🧑‍🔬 The cell theory (1838–1839) of Schleiden and Schwann says that...
 - a) plants have no cells
 - b) only animals have cells
 - c) life arises from mud
 - d) organisms are made up of cells
7. 💖 Darwin delayed publishing his theory for about 20 years because...
 - a) he had lost his notes
 - b) he could not write
 - c) he feared how society would react
 - d) he was waiting for the queen
8. 🍃 In which year did Charles Darwin publish "On the Origin of Species"?
 - a) 1859
 - b) 1831
 - c) 1882
 - d) 1809
9. ⭐ Darwin's theory of evolution caused great disputes because it held that...
 - a) the Earth is flat
 - b) humans and apes share common ancestors
 - c) breeding animals was wrong
 - d) there was life on Mars

CHEMISTRY

7. Industrial Revolution

1800–1900 AD



Let's step into a nineteenth-century laboratory. In 1808 John Dalton claimed that atoms of the same element are identical. In 1813 Jöns Jacob Berzelius introduced chemical symbols such as H, O and Fe. Friedrich August Kekulé said that he dreamed of a snake biting its own tail and thus discovered the ring structure of benzene. Robert Bunsen popularized a burner that gives a hot, blue flame. Helium was first discovered not on Earth but in the spectrum of the Sun. Alexander Borodin, composer of the opera "Prince Igor," was by profession a chemist. Alfred Nobel changed his will after reading a premature obituary that called him "the merchant of death." Marie Skłodowska went to Paris to study because in the Russian partition women could not attend university. In 1898, together with Pierre Curie, she discovered two elements: radium and polonium, named in honor of Poland. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Alexander Borodin, composer of the opera "Prince Igor," was by profession a...**
 - a) astronomer
 - b) blacksmith
 - c) chemist
 - d) ship's captain
2. 👁 **Friedrich August Kekulé dreamed of a snake biting its own tail and thus discovered...**
 - a) the periodic table
 - b) the hydrogen atom
 - c) the ring structure of benzene
 - d) the shape of DNA
3. 🗨 **The chemical symbols of the elements (e.g., H, O, Fe) were introduced in 1813 by...**
 - a) Dmitri Mendeleev
 - b) Antoine Lavoisier
 - c) Jöns Jacob Berzelius
 - d) Robert Boyle
4. ⚖ **John Dalton (1808) claimed that atoms of the same element...**
 - a) are alive
 - b) are identical
 - c) can be split in half
 - d) have different colors
5. 🔥 **Robert Bunsen popularized a burner that produces...**
 - a) cold fire
 - b) light without fire
 - c) a hot, blue flame
 - d) red smoke
6. 👤 **In 1898 Marie Skłodowska and Pierre Curie discovered two elements: polonium and...**
 - a) neon
 - b) radium
 - c) uranium
 - d) helium
7. ❤ **Marie Skłodowska went to Paris to study because in the Russian partition...**
 - a) women could not attend university
 - b) there were no books
 - c) the tsarist government banned chemistry
 - d) she did not know Russian
8. 🌿 **Helium was first discovered not on Earth but in...**
 - a) seawater
 - b) a volcano
 - c) a salt mine
 - d) the spectrum of the Sun
9. ☆ **Alfred Nobel changed his will after reading his own premature obituary, which called him...**
 - a) "the king of Sweden"
 - b) "the lord of inventions"
 - c) "the merchant of death"
 - d) "the father of peace"

PHYSICS

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, a century of great discoveries in physics. In the double-slit experiment Thomas Young showed that light behaves like a wave. In 1842 Christian Doppler explained why the siren of an approaching vehicle sounds higher, and of a receding one lower. The unit of energy, the joule, is named after James Joule, who showed that work can turn into heat. The first law of thermodynamics says that energy does not vanish but changes form. In 1851 Foucault's pendulum in the Paris Panthéon proved that the Earth rotates on its axis. Lord Kelvin introduced a temperature scale starting from absolute zero, which is about $-273\text{ }^{\circ}\text{C}$. In the 1860s James Clerk Maxwell showed that light is an electromagnetic wave. In 1897 Joseph Thomson discovered the electron. And Marie Skłodowska-Curie and Pierre Curie studied a phenomenon that Marie named radioactivity. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Christian Doppler (1842) explained why the siren of an approaching vehicle sounds...**
 - a) quieter up close
 - b) like an echo in the mountains
 - c) higher, and of a receding one lower
 - d) always the same
2. 👁 **In the double-slit experiment, Thomas Young showed that light behaves like...**
 - a) a liquid
 - b) sand
 - c) a wave
 - d) an electric current
3. 🗨 **The unit of energy, the joule, is named after...**
 - a) Blaise Pascal
 - b) Isaac Newton
 - c) James Watt
 - d) James Joule
4. ⚡ **In the 1860s James Clerk Maxwell showed that light is...**
 - a) a stream of fire
 - b) a sound
 - c) an electromagnetic wave
 - d) a gas
5. ⚙ **Foucault's pendulum in the Paris Panthéon (1851) proved that...**
 - a) time runs slower
 - b) the Earth rotates on its axis
 - c) air has mass
 - d) the Moon orbits the Earth
6. 👤 **Marie Skłodowska-Curie and Pierre Curie studied a phenomenon that Marie named...**
 - a) magnetism
 - b) radioactivity
 - c) electricity
 - d) gravity
7. ❤ **In 1897 Joseph Thomson discovered the smallest particle then known, the...**
 - a) electron
 - b) neutron
 - c) proton
 - d) quark
8. 🌿 **The first law of thermodynamics says that energy...**
 - a) arises from nothing
 - b) does not vanish but changes form
 - c) always runs out
 - d) exists only in fire
9. ☆ **In the 19th century Lord Kelvin introduced a temperature scale starting from absolute zero, which is...**
 - a) $-100\text{ }^{\circ}\text{C}$
 - b) $-1000\text{ }^{\circ}\text{C}$
 - c) $0\text{ }^{\circ}\text{C}$
 - d) about $-273\text{ }^{\circ}\text{C}$

ENVIRONMENT

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when industry began to change nature and people learned to protect it. In 1815 the Tambora volcano erupted, and its ash cooled the climate so much that 1816 was called "the year without a summer." Alexander von Humboldt described nature as one great web of connections. The poet William Wordsworth protested against building a railway in the Lake District. In 1866 the word "ecology" was coined by Ernst Haeckel. In 1868 the Galician Diet passed one of the first laws protecting chamois and marmots in the Tatras. In 1872 Yellowstone park was created, famous for the geyser Old Faithful. In 1883 the eruption of Krakatoa could be heard almost 5,000 km away. John Muir, who founded the Sierra Club in 1892, fought to protect the wilderness of the USA. And the passenger pigeon, once numbering in the billions, died out because it was shot in huge numbers. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 The poet William Wordsworth protested against building a railway in...
 - a) the Lake District
 - b) Scotland
 - c) Wales
 - d) London
2. 🗨️ Yellowstone, the world's first national park, is famous for the geyser...
 - a) Strokkur
 - b) Old Faithful
 - c) Wai-O-Tapu
 - d) Geysir
3. 📖 The word "ecology" was coined in 1866 by...
 - a) Alexander von Humboldt
 - b) Gregor Mendel
 - c) Charles Darwin
 - d) Ernst Haeckel
4. 🐦 The passenger pigeon, once numbering in the billions, died out at the turn of the 20th century because...
 - a) it froze in winter
 - b) cats ate it
 - c) it flew away to Africa
 - d) it was shot in huge numbers
5. 🌋 Because of the eruption of the Tambora volcano (1815), 1816 was called...
 - a) "the year of spring"
 - b) "the golden year"
 - c) "the year of fire"
 - d) "the year without a summer"
6. 🧑 John Muir, founder of the Sierra Club (1892), fought to protect...
 - a) railways
 - b) factories
 - c) the wilderness of the USA
 - d) gold mines
7. ❤️ Alexander von Humboldt described nature as...
 - a) the work of chance
 - b) one great web of connections
 - c) a collection of separate things
 - d) a steam engine
8. 🌿 The eruption of Krakatoa (1883) could be heard at a distance of...
 - a) 5 km
 - b) 50 km
 - c) almost 5,000 km
 - d) 500 meters
9. ⭐ In 1868 the Galician Diet passed one of the world's first laws protecting...
 - a) bison in Białowieża Forest
 - b) storks in Masuria
 - c) wolves in the Bieszczady
 - d) chamois and marmots in the Tatras

HEALTH

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when people were learning to fight disease. Ignaz Semmelweis ordered doctors to wash their hands in a chlorine solution before examining women, and deaths among mothers fell many times over. During the Crimean War Florence Nightingale saved the wounded by ensuring clean, well-ventilated hospitals. Cholera epidemics spread mainly through contaminated water, so water mains and sewers were built. The word "pasteurization" comes from the surname of Louis Pasteur, and in 1867 Joseph Lister introduced disinfection with carbolic acid into surgery. In 1882 Robert Koch discovered the bacterium that causes tuberculosis. Sebastian Kneipp hardened people by, among other things, walking barefoot on dew and in cold water. Also popular was preparing medicines from highly diluted doses, called homeopathy, though modern medicine does not confirm it works. And the chronically ill Chopin died in 1849 in Paris. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Chopin suffered from chronic illness and died in 1849 in...**
 - a) Warsaw
 - b) Paris
 - c) Vienna
 - d) London
2. 👁 **Before examining women, Ignaz Semmelweis ordered doctors to...**
 - a) wash their hands in a chlorine solution
 - b) burn herbs
 - c) wear wigs
 - d) pray
3. 🗨 **The word "pasteurization" comes from the surname of...**
 - a) Robert Koch
 - b) Joseph Lister
 - c) Edward Jenner
 - d) Louis Pasteur
4. ⚖ **In 1882 Robert Koch discovered the bacterium that causes...**
 - a) influenza
 - b) the common cold
 - c) tuberculosis
 - d) smallpox
5. ⚡ **Sebastian Kneipp treated people by, among other things, walking barefoot on...**
 - a) glowing coals
 - b) hot sand
 - c) ice
 - d) dew and in cold water
6. 🧑 **During the Crimean War Florence Nightingale improved the lot of the wounded by ensuring...**
 - a) bans on visits
 - b) clean, well-ventilated hospitals
 - c) new cannons
 - d) better uniforms
7. 💖 **In 1867 Joseph Lister introduced into surgery...**
 - a) operations by candlelight
 - b) ether anesthesia
 - c) disinfection with carbolic acid
 - d) treatment with leeches
8. 🍃 **In the 19th century cholera epidemics spread mainly through...**
 - a) mountain air
 - b) sunlight
 - c) contaminated water
 - d) mosquito bites
9. ☆ **In the 19th century, preparing medicines from highly diluted doses was called...**
 - a) hygiene
 - b) anatomy
 - c) homeopathy
 - d) surgery

PHYSICAL CULTURE AND SPORT

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when modern sport was born. In Norway, where skis had been used for centuries in winter, the first skiing competitions were held. In 1863 the English Football Association (FA) was founded, and football began to be played under written rules. In 1877 the first Wimbledon tennis tournament took place. In 1891 the teacher James Naismith invented basketball, and the first baskets were peach baskets. In 1895 William Morgan invented a less tiring game for businessmen: volleyball. At the first modern Games in 1896, women were not allowed to compete. The marathon was won by the Greek Spyridon Louis, and the Olympic anthem was composed for those Games by Spyridon Samaras. The motto "Citius, altius, fortius," adopted thanks to Coubertin, means faster, higher, stronger. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The Olympic anthem is a song composed for the 1896 Games by...**
 - a) Giuseppe Verdi
 - b) Richard Wagner
 - c) Spyridon Samaras
 - d) Frédéric Chopin
2. 🏈 **Football matches under written rules began in 1863 with the founding of the...**
 - a) Premier League
 - b) Real Madrid club
 - c) English Football Association (FA)
 - d) FIFA
3. 🏅 **The first Olympic marathon champion of the modern Games (1896) was the Greek...**
 - a) Alexandros Soutsos
 - b) Spyridon Louis
 - c) Demetrios Vikelas
 - d) Pierre de Coubertin
4. 🏀 **Basketball was invented in 1891 by the teacher James Naismith. The first baskets were...**
 - a) buckets
 - b) peach baskets
 - c) metal hoops
 - d) fishing nets
5. 🏏 **In 1895 William Morgan invented a less tiring game for businessmen:...**
 - a) hockey
 - b) tennis
 - c) golf
 - d) volleyball
6. 🎾 **The first Wimbledon tennis tournament was held in...**
 - a) 1800
 - b) 1850
 - c) 1900
 - d) 1877
7. ❤️ **The Olympic motto "Citius, altius, fortius," adopted thanks to Coubertin, means...**
 - a) faster, higher, stronger
 - b) win at any cost
 - c) peace, sport, friendship
 - d) together, further, better
8. 🏂 **The first skiing competitions were organized in the 19th century in...**
 - a) Brazil
 - b) Egypt
 - c) Australia
 - d) Norway
9. ⭐ **At the first modern Games in 1896, the athletes not allowed to compete were...**
 - a) Greeks
 - b) women
 - c) students
 - d) Americans

MEDICINE

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when medicine took giant steps. Laennec's first stethoscope was a wooden tube resembling a flute. Edward Jenner opened the way to vaccines in 1796, and the word "vaccine" comes from the Latin "vacca," meaning cow. In 1846 the dentist William Morton administered ether anesthesia during an operation. In 1849 Elizabeth Blackwell became the first woman in the USA to earn a medical degree. In 1853, during childbirth, Queen Victoria popularized chloroform anesthesia. In 1858 "Gray's Anatomy" appeared, a famous textbook with drawings of the body's structure. The red cross on a white background is the reversed flag of Switzerland. In 1885 Louis Pasteur saved a boy bitten by a rabid dog by giving him a rabies vaccine. And in 1897 Ronald Ross showed that malaria is transmitted by mosquitoes. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎺 **Laennec's first stethoscope was a wooden tube resembling a...**
 - a) trumpet
 - b) lyre
 - c) drum
 - d) flute
2. 👁 **"Gray's Anatomy" (1858) is a famous textbook with...**
 - a) drawings of the body's structure
 - b) sheet music
 - c) maps of the world
 - d) recipes
3. 🗨 **Edward Jenner (1796) created the smallpox vaccine, and the word "vaccine" comes from the Latin "vacca," meaning...**
 - a) cow
 - b) needle
 - c) medicine
 - d) pox
4. 🐕 **In 1885 Louis Pasteur saved a boy bitten by a rabid dog by giving him...**
 - a) a rabies vaccine
 - b) an antibiotic
 - c) lemon juice
 - d) a hot bath
5. ⚡ **The first ether anesthesia during an operation (1846) was administered by the dentist...**
 - a) Charles Darwin
 - b) Robert Koch
 - c) William Morton
 - d) Louis Pasteur
6. 🏆 **In 1849 Elizabeth Blackwell became the first woman in the USA to...**
 - a) win an Olympic medal
 - b) invent the light bulb
 - c) become president
 - d) earn a medical degree
7. 💖 **During childbirth in 1853, Queen Victoria popularized...**
 - a) chloroform anesthesia
 - b) massage
 - c) vaccination
 - d) the cesarean section
8. 🍀 **In 1897 Ronald Ross showed that malaria is transmitted by...**
 - a) birds
 - b) mosquitoes
 - c) rats
 - d) fish
9. ⭐ **The red cross on a white background, a symbol of aid to the wounded, is the reversed flag of...**
 - a) Poland
 - b) Switzerland
 - c) Denmark
 - d) England

PHARMACY

7. Industrial Revolution

1800–1900 AD



Let's step into a nineteenth-century pharmacy. In 1804 the pharmacist Friedrich Sertürner isolated morphine from opium. Quinine, a malaria drug, was obtained from the bark of the cinchona tree. Drugs from the bark, leaves and roots of plants are studied by pharmacognosy. Pills were made by hand, rolling the mass on a pill machine, or pill roller, and poisons were kept in bottles of special shapes and colors to prevent mistakes. Working in a pharmacy in Lviv, the pharmacist Ignacy Łukasiewicz, together with Jan Zeh, refined kerosene from crude oil. Cocaine was used as an anesthetic in eye surgery before its dangers became known. In 1886 the pharmacist John Pemberton invented Coca-Cola as a medicinal drink, a tonic. And in 1899 aspirin appeared, with acetylsalicylic acid derived from a substance in willow bark. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚫 **Coca-Cola (1886) was invented by a pharmacist as...**
 - a) a medicinal drink (tonic)
 - b) rat poison
 - c) a varnish
 - d) a perfume
2. 👁 **Morphine was isolated from opium in 1804 by the pharmacist...**
 - a) Joseph Lister
 - b) Louis Pasteur
 - c) Robert Koch
 - d) Friedrich Sertürner
3. 📖 **Quinine, a malaria drug, was obtained from the bark of the...**
 - a) birch
 - b) cinchona tree
 - c) oak
 - d) pine
4. ➡ **Aspirin (1899) contains acetylsalicylic acid, derived from a substance found in the bark of...**
 - a) pine
 - b) willow
 - c) apple trees
 - d) maple
5. ⚙ **19th-century pharmacies made pills by hand, rolling the mass on a...**
 - a) pill machine (pill roller)
 - b) sewing machine
 - c) lathe
 - d) loom
6. 👤 **Working in a pharmacy in Lviv, the pharmacist Ignacy Łukasiewicz, together with Jan Zeh,...**
 - a) refined kerosene from crude oil
 - b) invented aspirin
 - c) discovered penicillin
 - d) created a vaccine
7. 💖 **In the 19th century cocaine was used as...**
 - a) a dye
 - b) a fertilizer
 - c) an anesthetic in eye surgery
 - d) a drink for children
8. 🌿 **Drugs from the bark, leaves and roots of plants are studied by...**
 - a) astronomy
 - b) pharmacognosy
 - c) geology
 - d) mechanics
9. ⭐ **To prevent mistakes, 19th-century pharmacies kept poisons in bottles...**
 - a) of special shapes and colors
 - b) without labels
 - c) in a neighbor's cellar
 - d) mixed with sugar

AGRICULTURE

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when machines rolled into the countryside. In 1826 the first beet sugar factory in the Kingdom of Poland was founded in Częstocice. In 1831 Cyrus McCormick built a mechanical reaper, and in 1837 John Deere made a plow with a steel share that did not get clogged with heavy prairie soil. Steam threshers roared in the fields, powered by a portable steam engine on wheels. Justus von Liebig showed that plants need minerals such as nitrogen and phosphorus, and guano from seabird droppings was imported from Peru. Potatoes became the basis of the daily diet of poor families, which is why the potato blight, caused by a fungus-like microorganism, was so dangerous. And the American plant breeder Luther Burbank developed the potato variety Russet Burbank. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The first steam threshers in the fields could be heard from afar; they were powered by...
 - a) a windmill
 - b) a portable steam engine on wheels
 - c) a horse on a treadmill
 - d) electricity from the grid
2. 👁 In 1831 Cyrus McCormick built a mechanical...
 - a) lawn mower
 - b) reaper
 - c) milking machine
 - d) watering can
3. 🗨 Justus von Liebig showed that plants need...
 - a) heat from a stove
 - b) music
 - c) only moonlight
 - d) minerals such as nitrogen and phosphorus
4. 🚢 Guano, a fertilizer made from seabird droppings, was imported to Europe from...
 - a) Poland
 - b) Norway
 - c) China
 - d) Peru
5. ⚡ In 1837 John Deere made a plow with a steel share that...
 - a) plowed the field by itself
 - b) planted grain
 - c) did not get clogged with heavy prairie soil
 - d) was powered by electricity
6. 🧑 The first beet sugar factory in the Kingdom of Poland was founded in...
 - a) Warsaw (1900)
 - b) Częstocice (1826)
 - c) Kraków (1700)
 - d) Gdańsk (1500)
7. ❤ In the 19th century, in many European countries potatoes became the basis of...
 - a) fodder for horses only
 - b) paper production
 - c) the daily diet of poor families
 - d) royal cuisine
8. 🌱 Luther Burbank, an American plant breeder, developed the potato variety...
 - a) Irish Red
 - b) Kraków Gold
 - c) Royal Spud
 - d) Russet Burbank
9. ☆ The potato blight that caused the famine in Ireland is a disease caused by...
 - a) cold rains
 - b) a fungus-like microorganism
 - c) insects from America
 - d) mosquitoes

VETERINARY SCIENCE

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, when animals still worked side by side with people. Horses pulled trams, omnibuses and cabs, and the clatter of hooves was a typical sound of cities. Blacksmiths and veterinarians cared for hooves by nailing on horseshoes. The world's first veterinary school had been founded back in 1762 in Lyon, and in the nineteenth century such schools operated all over Europe. In 1824 the first animal protection society, the RSPCA, was founded in England. In 1877, in "Black Beauty," Anna Sewell told the life of a horse from its own point of view. In 1878 Eadweard Muybridge used a series of photographs to check whether a galloping horse lifts all its hooves off the ground – and it does. In 1881 Louis Pasteur developed a vaccine against anthrax. In the 1890s a cattle plague in Africa killed millions of cattle and antelopes. And the faithful dog Greyfriars Bobby kept watch at his master's grave for 14 years. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the 19th century horses pulled trams, omnibuses and cabs; the clatter of their hooves was...
 - a) a typical sound of cities
 - b) heard only in the countryside
 - c) a rarity
 - d) banned at night
2. 📷 In 1878 Eadweard Muybridge used a series of photographs to check whether a galloping horse...
 - a) has four legs
 - b) can jump
 - c) lifts all its hooves off the ground
 - d) is faster than a train
3. 📖 The world's first veterinary school was founded in 1762 in...
 - a) Warsaw
 - b) Boston
 - c) Lyon
 - d) Rome
4. 🦠 In 1881 Louis Pasteur developed a vaccine for sheep and cattle against...
 - a) influenza
 - b) anthrax
 - c) the common cold
 - d) scabies
5. 🛠 In the 19th century blacksmiths and veterinarians cared for horses' hooves by nailing on...
 - a) socks
 - b) spurs
 - c) little bells
 - d) horseshoes
6. 🐾 The first animal protection society (RSPCA) was founded in 1824 in...
 - a) Russia
 - b) Poland
 - c) England
 - d) Brazil
7. 💖 In the novel "Black Beauty" (1877) Anna Sewell told the life of a horse...
 - a) from its own point of view
 - b) in the form of a poem
 - c) from a king's point of view
 - d) as a war chronicle
8. 🌿 The cattle plague in Africa (1890s) killed millions of...
 - a) fish
 - b) birds
 - c) cattle and antelopes
 - d) elephants
9. ⭐ For 14 years the faithful dog Greyfriars Bobby of Edinburgh...
 - a) drove a train
 - b) guarded a castle
 - c) kept watch at his master's grave
 - d) rescued drowning people



ANIMAL SCIENCE AND FISHERIES

7. Industrial Revolution

1800–1900 AD

Let's travel back to the nineteenth century, when animal breeding and fishing changed along with industry. Whalers risked their lives because they hunted from boats with hand harpoons, and while working they sang sea shanties. Whale oil was used mainly for lighting lamps until kerosene replaced it. Breeders developed selective animal breeding, started by Robert Bakewell. The monk Gregor Mendel, before turning to peas, tried to study heredity in mice. In Germany a dog was bred to guard flocks – we call this breed the German Shepherd. In the Wild West cowboys drove cattle to railway stations. North American bison nearly died out because they were hunted on a massive scale. The first artificial fish hatcheries also appeared, to restock rivers with young fish. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **19th-century whalers sang songs called...**
 - a) sea shanties
 - b) marches
 - c) psalms
 - d) Christmas carols
2. 👁 **In the 19th century whale oil was used mainly for...**
 - a) lighting lamps
 - b) painting pictures
 - c) cooking soups
 - d) drinking
3. 🐕 **The dog breed developed in 19th-century Germany to guard flocks is called the...**
 - a) Chihuahua
 - b) German Shepherd
 - c) Poodle
 - d) Husky
4. 🧬 **Robert Bakewell started in the 18th century, and 19th-century breeders developed,...**
 - a) selective animal breeding
 - b) silkworm farming
 - c) dinosaur breeding
 - d) whaling
5. 🐮 **Cowboys in the Wild West drove cattle to...**
 - a) warships
 - b) pastures in Europe
 - c) markets in China
 - d) railway stations
6. 🐟 **The first artificial fish hatcheries were set up in the 19th century to...**
 - a) catch whales
 - b) feed cats
 - c) restock rivers with young fish
 - d) make plastic caviar
7. 💚 **Before turning to peas, the monk Gregor Mendel tried to study heredity in...**
 - a) mice
 - b) chickens
 - c) bees
 - d) horses
8. 🌿 **In the 19th century North American bison nearly died out because...**
 - a) wolves ate them
 - b) they froze
 - c) they were hunted on a massive scale
 - d) starving cowboys ate them
9. ⭐ **Whalers risked their lives because...**
 - a) whales attacked ports and towns
 - b) they had to dive without any gear
 - c) they hunted from boats with hand harpoons
 - d) they hunted only on dark nights

1.a.2.a.3.b.4.a.5.d.6.c.7.a.8.c.9.c

A N S W E R S

OTHER (NATURE)

7. Industrial Revolution

1800–1900 AD



Let's travel back to the nineteenth century, a century of explorers and travelers. Early in the century Georges Cuvier, comparing the mammoth with elephants, proved that some species have gone extinct. Mary Anning, a self-taught woman from England, discovered in the cliffs skeletons of ichthyosaurs and plesiosaurs. In 1842 Richard Owen coined the word "dinosaur," meaning terrible lizard. In 1802 Alexander von Humboldt tried to climb the volcano Chimborazo. In 1840 Paweł Edmund Strzelecki named Australia's highest peak after Kościuszko. The Polish exiles Jan Czerski and Aleksander Czekanowski explored Siberia, and Ignacy Domeyko, an exile from Lithuania, became a national hero of Chile. Charles Darwin believed that birdsong serves mainly to attract mates. And Robert Peary and other polar explorers tried to reach the North Pole by dog sled. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 📖 **Charles Darwin believed that birdsong serves mainly to...**
 - a) frighten people
 - b) announce rain
 - c) keep track of time
 - d) attract mates
2. 👁 **Mary Anning, a self-taught woman from England, discovered in the cliffs...**
 - a) the buried ruins of ancient Troy
 - b) veins of gold and silver ore
 - c) skeletons of ichthyosaurs and plesiosaurs
 - d) fallen iron meteorites
3. 🗨 **The word "dinosaur" ("terrible lizard") was coined in 1842 by...**
 - a) Richard Owen
 - b) Mary Anning
 - c) Charles Darwin
 - d) Georges Cuvier
4. ⚖ **At the beginning of the 19th century Georges Cuvier proved that some species...**
 - a) arise from mud
 - b) have gone extinct
 - c) live forever
 - d) turn into stone
5. 🧭 **In the 19th century Robert Peary and other polar explorers tried to reach the North Pole...**
 - a) by submarine
 - b) by dog sled
 - c) by airplane
 - d) by rail
6. 🏔 **Paweł Edmund Strzelecki named Australia's highest peak after...**
 - a) Piłsudski
 - b) Copernicus
 - c) Kościuszko
 - d) Chopin
7. ❤ **In 1802 Alexander von Humboldt tried to climb the volcano...**
 - a) Vesuvius
 - b) Etna
 - c) Fuji
 - d) Chimborazo
8. 🌿 **Jan Czerski and Aleksander Czekanowski, Polish exiles, explored...**
 - a) Africa
 - b) Siberia
 - c) Antarctica
 - d) the Amazon
9. ☆ **Ignacy Domeyko, an exile from Lithuania, became a national hero of...**
 - a) Chile
 - b) Canada
 - c) Egypt
 - d) Japan

1.d 2.c 3.a 4.b 5.b 6.c 7.d 8.b 9.a

A N S W E R S